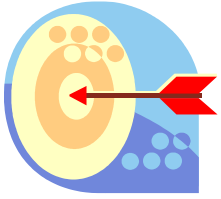


Lesson 11: Tracking System Health

Objectives



This lesson explains how to locate and correct potential problems in a business process. For instances that stopped running, you must troubleshoot the instance to identify the cause, reconcile the issue or error, and determine whether to restart, resume, or terminate the instance.

This lesson will cover:

- Troubleshooting and reconciling business processes
- Archiving business process data
- Restoring business process data archived offline
- Managing Resources
- Deleting a Business Process

Troubleshoot and Reconcile Business Processes

- Sterling B2B Integrator has a built-in system recovery business process that runs automatically every 45 minutes. The recovery business process monitors active business process instances and allows Sterling B2B Integrator to report any issues that occur: for example, a system or service associated with an instance shuts down or a compliance error results while an instance that implements standards is running.
- When the recovery business process identifies issues or errors while an instance is running, Sterling B2B Integrator halts the instance, indicates an advance status message of error and a status message of interrupted, and generates reports, as appropriate.

Troubleshoot and Reconcile Business Processes

When troubleshooting an instance that indicates a status error, remember that:

- The terminated activity and state activity override all other activities and states.
- When an instance is terminated, all subprocesses and any subprocesses waiting to be processed in the instance terminate.

Used in combination, the status indicator, advanced status and state messages, and reports can help to determine:

- Issues that result while an instance is running
- Errors or warnings that prevent an instance from running
- Actions to complete to reconcile errors or warnings

Troubleshooting and Reconciling Business Processes (continued)



Please pause this presentation and review the table that describes the status indicators and state messages and possible actions to take in response to them that are listed in your student manual. There is also a section on Advanced Status Messages.

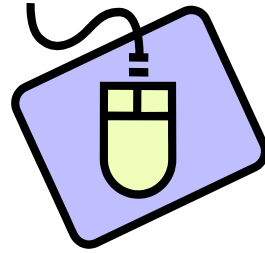
Troubleshooting and Reconciling Business Processes (continued)



Viewing Troubleshooting Reports:

Sterling B2B Integrator generates two reports to use for troubleshooting business processes. The reports are:

- Status Report
- EDI Compliance



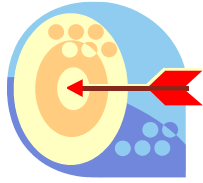
Exercise 11–1: Viewing a Troubleshooting Report

Your turn!

Pause this presentation, view a troubleshooting report by following the steps listed in your student manual in exercise 11-1

Archiving and Restoring Business Process Data

Archiving and Restoring Business Process Data



Archiving your business process data not only protects your critical data but also conserves database disk space, improving the efficiency of Sterling B2B Integrator.

Sterling B2B Integrator uses a set of components to move business process data out of the database and onto a persistence level of long-term storage.

The components include:

- Index Business Process service
- Archive system business process
- Purge system business process

Data that has been removed using the Backup Business Process component can later be restored to the same version of Sterling B2B Integrator using the Restore Business Process component.

Database and Persistence Level



Important!!!

The level of detail written to the database as a business process runs is affected by the persistence level, which you specify with the Business Process Manager or in the noapp.properties file.

Decreasing the persistence level increases the business process performance at the cost of full-tracking for each step of the business process.

Archiving and Restoring Business Process Data (continued)

Business processes are eligible for archiving as soon as their archive flag is set by the Index Business Process service. You do not have to wait until a business process exceeds its life span before archiving it. After the archive flag is set, the business process is then archived the next time the Backup Business Process service is run either by schedule (at 2 A.M. each morning) or manually. The data is still available in the system because it cannot be purged (with the Purge service) until it has exceeded its life span.

The Restore Business Process service restores archived business process data from physical media to a Sterling B2B Integrator restored data location where it can be searched and viewed.

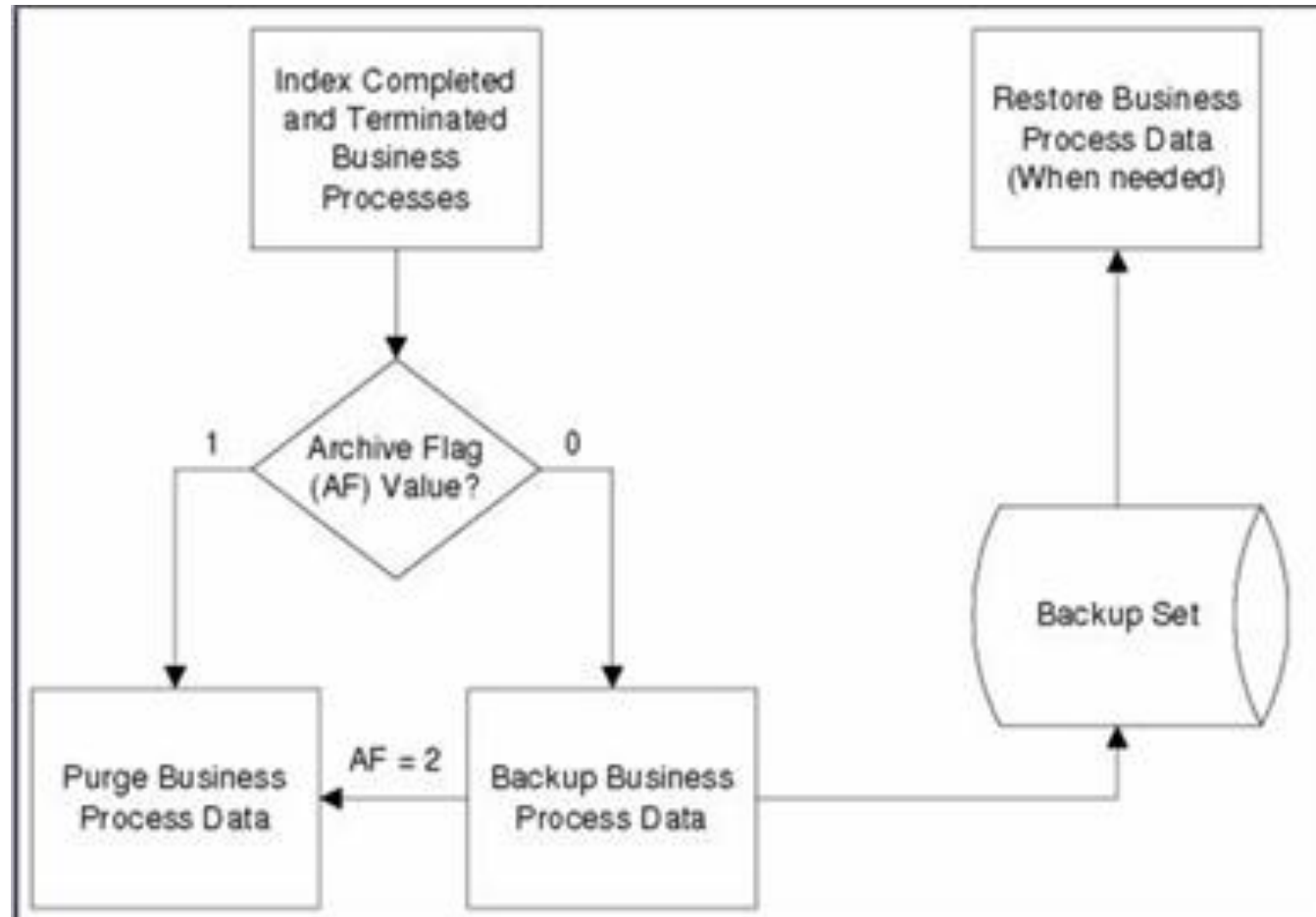
Archiving and Restoring Business Process Data (continued)

After running a business process, when a business process terminates, and before archiving or purging business process data, the business process must first be indexed. That is, the Index Business Process service runs a process that looks for archiving parameters defined in the business process. When the process finds these parameters, the process creates summary information and writes the information to a table in the database.

For information written to the table, the Backup Business Process service runs, according to how you have configured your archiving settings. The Backup Business Process service retains the business process and its related data to a file system as Java™ serialized objects for later backup to tape, some other long-term storage medium, or purging. The Purge service then runs and removes any data from the database or file system, or both, specified in archive settings not to retain.

Archiving and Restoring Business Process Data (continued)

The following diagram shows the archiving process.



Archiving and Restoring Business Process Data (continued)



Important!!!!

You cannot archive active business process instances (for example, halting, halted or interrupted). Only data for completed or terminated business processes that have been indexed can be archived.

You cannot delete a workflow definition of a business process (using the Source Manager of the Business Process Manager) while an instance of that process is being either indexed, archived or purged. If you try to delete such a definition, an error message appears.

Factors affecting the Amount of Database Table Space

There are two options that affect the amount of database table space used when you archive or purge business processes:

- **Persistence level** – Amount of information stored when a BP runs.
- **Life span** – How long information stays in the database after a BP runs.

Persistence levels – Here are some sample persistence levels available (See Knowledge Center or your Student Guide for more info)

- **Full** - Retain all data for this business process, including associated documents, activities, and all process data.
- **Step Status (Engine May Override)** - This is the recommended persistence level for most process models because it helps optimize performance while providing a level of recovery suitable for most business needs. Persists process data and documents for service steps according to the persistence level supported by the service in the step.
- **BP Start Stop (Engine May Override)** - Choose this level if you know that you do not need persisted status data for successful processing. Because service level settings override the selection, you can have additional data in a recovery scenario. Persists process data and documents for service steps according to the persistence level supported by the service in the step.
- **Zero** - This level does not persist any business process data for recovery or process tracking. Choose this level only if you are certain you will not need the data.
- **Error Only** - Choose this level if you do not need tracking data for your process when it is successful and your recovery does not depend on process step data being saved.

Life span

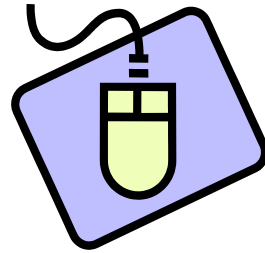
- This option allows you to define the length of time, in days and hours, to retain the data in Sterling B2B Integrator ONLINE tables. This is the length of time after which the business process instance expires.

Configuring Archive Settings

You can configure Sterling B2B Integrator to automatically archive business process data as often as you need.

When archiving your business process data, consider that:

- You cannot archive active business process instances. Only data for completed or terminated business processes can be archived.
- You should indicate default values that apply to all business process instances that you run.
- How often you archive should depend on the volume of business processes that you run.
- Business level information should be archived for audit purposes, including:
 - Security information, such as certificates
 - Documents that support non-repudiation for AS1 and AS2 protocols
 - Correlations that provide searching ability
- Business process are purged only when they have exceeded their life span.
- For any data that you may potential want to view in the immediate future, maintaining the data in live tables provides faster retrieval.



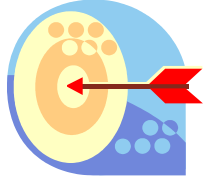
Exercise 11–2: Archiving

Your turn!

Pause this presentation, and follow the instructions for configuring default archive settings in your student manual.

Initiating Archiving

Initiating Archiving



After you have configured archive settings, you can initiate archiving for all completed or terminated business process instances at any time.

To initiate archiving:

1. From the Sterling B2B Integrator Operations menu, select Archive Manager.
2. In the Archive Execution section, click Go! to run the archiving process.

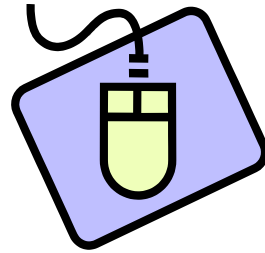
The only archived business process data that you can view from its archived location is data archived to archive tables within Sterling B2B Integrator. To view archived business process data that has been archived to an offline location, you must first restore that data within Sterling B2B Integrator.

Initiating Archiving (continued)

In Sterling B2B Integrator, you can view the archive log using either the:

- Archive Manager
- System logs

After you have archived business process data to an offline location, you can restore that data and view it in Sterling B2B Integrator.



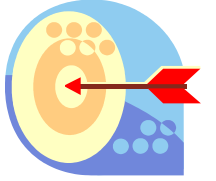
Exercise 11–3: Restoring Archived Business Process Data

Your turn!

Pause this presentation, and follow the instructions on restoring an archived business process. And the steps on how to view restored business process data.

Managing Resources

Managing Resources



In Sterling B2B Integrator, resources are files, templates, and documents that are deployed in Sterling B2B Integrator to perform various actions. Sterling B2B Integrator enables you to import and export resources, which can save time and increase the accuracy of duplicating resources on various Sterling B2B Integrator systems that are set up for unique purposes.

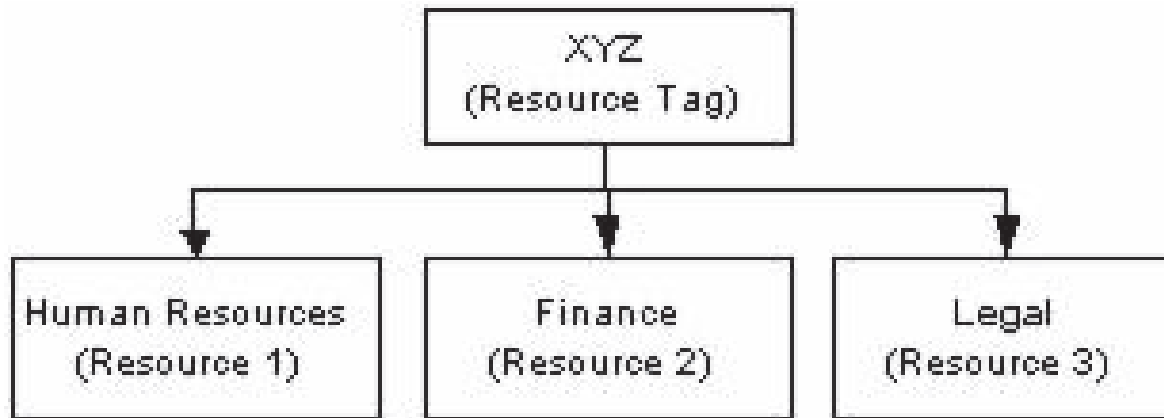
Specifically, the Import and Export options enable you to:

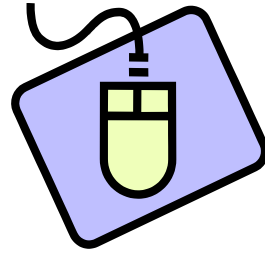
- Import resources from an Sterling B2B Integrator test environment to a Sterling B2B Integrator production environment.
- Import resources from one Sterling B2B Integrator system to another.

To import and export resources from Sterling B2B Integrator environment to another Sterling B2B Integrator environment, both environments must be the same version.

Resource Tags

- A resource tag is a name that you use to identify a group of associated resources. A resource tag is much like a company name, with each department being a web resource associated with the resource tag. For example, XYZ company has Human Resources, Finance, and Legal departments. The company name, XYZ, is the resource tag, and the Human Resources, Finance, and Legal departments are web resources that are associated with the XYZ resource tag.
- Sterling B2B Integrator allows you to associate a resource with more than one resource tag.





Exercise 11–4: Creating Resource Tags

Your turn!

Pause this presentation, and follow the instructions on Creating Resource Tags

Exporting Resources

Exporting Resources

Before conversion, resources must be converted into the proper format for storage and transference. The Export option converts a group of resources that you specify into one of the following formats:

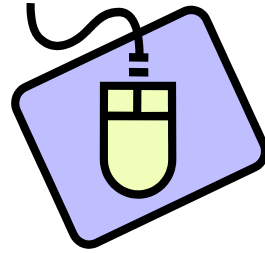
- XML - Enables you to transfer data or resources between two existing systems.
- Install bundle - Optionally enables you to load during an Sterling B2B Integrator install on a new system.

In the export process, after you have defined the file format to use for the export, you select the version of the resources to export:

- The Standard export copies non-versioned resources and the default version of versioned resources.
- The Advanced export copies non-versioned resources and enables you to choose to export just default versions or all versions of versioned resources.

You can export resources that are associated with a particular resource tag or resources not associated with a resource tag.

Important!!! You cannot use the Export option to export .war files.



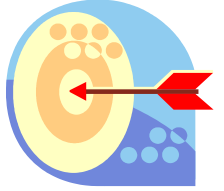
Exercise 11–5: Exporting Resources

Your turn!

Pause this presentation, and follow the instructions on Exporting Resources

Importing Resources

Importing Resources



When you import resources, the Import option converts an XML file or install bundle to Sterling B2B Integrator resources format.

Depending on the type of export you used (standard or advanced, default or all versions), the Import option performs the following functions:

- Creates new non-versioned resources
- Creates and checks in new versioned resources (assigns time/date of the import)
- Updates or preserves existing non-versioned resources
- Preserves or appends existing checked-in resources

**Caution**

To prevent the loss or corruption of existing records, preserve and download the backup file of all imports that you conduct.

Importing Resources

(Continued)

Here is some important information about importing resources:

You must manually update imported resources that contain hard-coded, computer-specific information. For example, system paths in scripts must be manually changed to use the new path after importing resources to their new location.

During import, Sterling B2B Integrator creates a backup file containing records as they existed prior to import. After you import the resources, you can download and preserve the backup file, which is named backup.xml by default. You can change the name of the backup file so that you do not overwrite an existing backup file.

For both non-versioned and versioned resources, the import process creates a new record where none exists. If you are importing resources to an environment that already contains the same resources, you have two options:

- Update
- Preserve Existing (default)

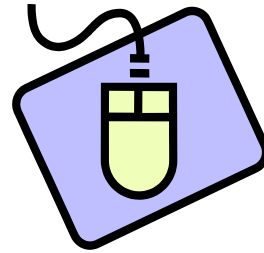
Importing Resources

(Continued)

The Preserving Existing option is the default option. Depending on whether you are working with versioned or non-versioned resources, different actions occur.

- Non-versioned resources - If a record already exists, nothing is imported. The import process does not change existing records.
- For versioned resources - The preserve existing option does not change any versions currently in the system, and adds only new versions.

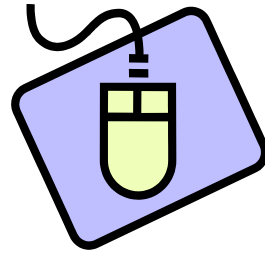
You see the Update option only when you select Update during the import process. If you are importing non-versioned resources, the Update option replaces the existing records. If you are importing versioned resources, the Update option appends to the list when versions already exist, incrementing by one, according to the starting point of the import system.



Exercise 11–6: Importing Resources into Sterling B2B Integrator

Your turn!

Pause this presentation, and follow the instructions on adding information into Sterling B2B Integrator.



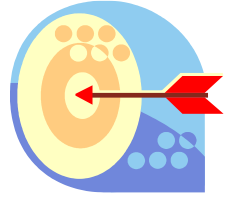
Exercise 11–7: Importing and Exporting Using Resource Tags

Your turn!

Pause this presentation, and follow the instructions on Importing and Exporting Using Resource Tags.

Deleting a Business Process

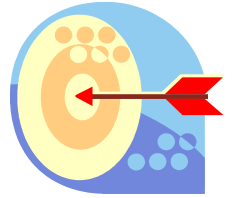
Deleting a Business Process



Overview: In Sterling B2B Integrator, you can delete individual business processes (except business processes used by Sterling B2B Integrator). Some reasons you might delete a business process are:

- Business activities have changed and, as a result, a business process can no longer accomplish the activities.
- You need to remove versions of a business process to reduce the database size.

Deleting a Business Process (continued)



Before deleting a business process, consider that:

- Deleting a business process is permanent; it cannot be restored once deleted.
- If the business process runs on a schedule, the schedule is deleted with the business process.
- When a user account is required by a business process (for example, to perform a human interaction activity, such as approving a PO), and that user account is deleted, the business process does not run, and Sterling B2B Integrator generates an error message instead.
- After you confirm that you want to delete a business process, Sterling B2B Integrator deletes the business process and creates a log entry that contains the name of the business process, date and time of deletion, and the user account who performed the deletion.
- After a service configuration has been deleted, you can reuse the name.
- Only user accounts that have SII_DELETE and BPMANAGE permissions can delete business processes.

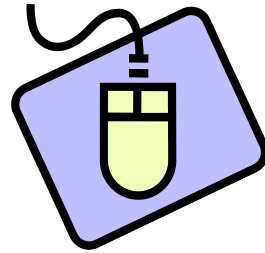
Deleting a Business Process

(continued)



Before you delete a business process, ensure that you have completed the following tasks:

1. Use the Export Resources function to save a copy of the business process to offline storage.
2. Identify any uses of the business process to ensure the deletion does not prevent other business processes and ensuing services from running.



Exercise 11–8: Deleting a Business Process

Your turn!

*Pause this presentation, and follow the instructions on
Deleting a Business Process.*

Knowledge Check



Please Pause this presentation and complete Quiz in your Student Manual.

Lesson Review



Congratulations !!!!

In this lesson we covered the following:

- Troubleshooting and reconciling business processes
- Archiving business process data
- Restoring business process data archived offline
- Managing Resources
- Deleting a Business Process